Problems on Infinite Series – Comparison test — with Solutions

Problems:

Test the convergence of the following series:

1. $1-\frac{1}{3}+\frac{1}{3^2}-\frac{1}{3^3}+\cdots$
2. $\frac{3}{5}+\frac{4}{5^2}+\frac{3}{5^3}+\frac{4}{5^4}+\cdots$
3. $\frac{1}{1\cdot 2}+\frac{2}{3\cdot 4}+\frac{3}{5\cdot 6}+\cdots$
4. $\frac{1}{1\cdot 3}+\frac{2}{3\cdot 5}+\frac{3}{5\cdot 7}+\cdots$
5. $\frac{2}{1^p}+\frac{3}{2^p}+\frac{4}{3^p}+\frac{5}{4^p}+\cdots$
6. $\frac{1}{1^2}+\frac{1+2}{1^2+2^2}+\frac{1+2+3}{1^2+2^2+3^2}+\cdots$
7. $\frac{1}{1\cdot 3\cdot 5}+\frac{2}{3\cdot 5\cdot 7}+\frac{3}{5\cdot 7\cdot 9}+\cdots$
8. $\frac{3}{1}+\frac{4}{8}+\frac{5}{27}+\frac{6}{64}+\cdots$
9. $\sum_{n=1}^{\infty}\frac{\sqrt{n}}{n^2+1}$
10. $\sum_{n=0}^{\infty}\frac{2n^3+5}{4n^5+1}$
11. $\sum_{n=1}^{\infty}\frac{(n+1)(n+2)}{n^2\sqrt{n}}$
12. $\sum_{n=1}^{\infty}\left(\sqrt{n^2+1}-n\right)$
13. $\sum_{n=1}^{\infty}\left(\sqrt[3]{n^3+1}-n\right)$
14. $\sum_{n=1}^{\infty}\left(\sqrt{n^4+1}-\sqrt{n^4-1}\right)$
15. $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}\sin\left(\frac{1}{n}\right)$
16. $\sum_{n=2}^{\infty}\frac{1}{\sqrt{n}\text{lo}}$
17. $\sum_{n=1}^{\infty}\frac{\sqrt{n+1}-1}{(n+2)^3-1}$